Youngsam Chun

Responsible AI Researcher · AI Safety, Governance, and Value Alignment

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Profile

AI safety and governance researcher with applied industry experience at KT Corporation. My work spans red-teaming and defence of retrieval-augmented generation systems, responsible AI governance and risk taxonomies for high-risk and Physical AI systems, and value alignment and evaluation of large language model judges. I combine rigorous evaluation methods (adversarial testing, alignment auditing and structured risk taxonomy design) with policy-facing governance frameworks, and I aim to advance AI safety, fairness and governance research within the UK's AI governance research community.

Research Areas

- AI safety, red-teaming, and adversarial evaluation of LLM and RAG systems
- Responsible AI governance and risk taxonomies for high-risk and Physical AI systems
- Value alignment and evaluation of LLM judges (moral orientation and calibration)
- Agentic RAG security, black-box poisoning attacks, and defences
- Physical AI risk, embodied AI, and cyber-physical system safety
- Patent analytics, scientometrics, and AI innovation measurement
- Network science, economic complexity, and knowledge-space modelling

Education and Training

PhD, Technology Management, Economics, and Policy (TEMEP)
Mar 2019 – Aug 2024
Seoul National University, Seoul, Korea — Final grade 4.28/4.5 (97.8); EQF 8
Thesis: *Exploring complementary innovation of AI technology from a multidimensional knowledge network approach* (HAL: tel-05047941)

MSc, Technology Management, Economics, and Policy (TEMEP)
Mar 2017 – Feb 2019
Seoul National University, Seoul, Korea — Final grade 4.33/4.5 (98.3); EQF 7
Thesis: *Heterophily Effects on Industrial Innovation Leadership Changes in Autonomous Vehicle Industry*

Economic Networks Summer School
Jun 2024
University of Oxford, Oxford, United Kingdom · complex network analysis, economic complexity and geography.

Masterclass: MSCA Post-Doctoral Fellowships
May 2025
Université Paris 1 Panthéon-Sorbonne, Paris, France · intensive proposal-writing workshop.

Work Experience

Responsible AI Researcher · Responsible AI Department, Frontier AI Lab, KT Corporation
Aug 2024 – Present
Seoul, Korea

- Establishment of a responsible AI governance framework for high-risk domains.
- Joint research and development of a Korea-specific custom GPT-4o with Microsoft.
- Development of value-alignment technology reflecting Korea's cultural context.

Adjunct Professor in Business Analytics · UNIST
Aug 2024 – Present
Graduate School of Technology and Innovation Management, Ulsan National Institute of Science and Technology, Ulsan, Korea

- Graduate teaching on business analytics and machine/deep learning using Python.
- Supervision of graduate students in thesis writing and research.

Publications and Preprints

KidnapRAG: A Black-Box Attack for Hijacking Reasoning in Agentic Retrieval-Augmented Generation Systems 2026

Choi, C., Kim, E., Lee, K., **Chun, Y.**, Jeong, J., Kim, E., Oh, M., Jang, J., & Chang, B. (2026). *Proceedings of EMNLP 2026*, Budapest **(accepted, Main Conference; 15.4% acceptance rate)**. Industry-academia collaboration with Prof. Buru Chang's group (Korea University); **Chun, Y.** led the KT side of the project, is fourth author on the paper and holds first-inventor priority on the registered Korean patent (Jul 2026). arXiv:2607.00422 · OpenReview · Code

ARS-Bench: Benchmarking Security Risks Across the Agentic RAG Attack Lifecycle 2026

Choi, C., Kim, E., Lee, K., **Chun, Y.**, Jeong, J., Kim, E., Oh, M., Jang, J., & Chang, B. (2026). *Under review, ACL Rolling Review* (August 2026). A benchmark spanning the full agentic RAG attack lifecycle, from adversarial document ingestion to final response generation, structured along five security-critical dimensions.

Silent Failure of Moral Alignment in Large Language-Model Evaluators 2026

Chun, Y. *Under review at Scientific Reports* (June 2026). Reproducibility release: deep1003/silent-failure-moral-alignment.

Moral Orientation and Calibration: Coupled in Human Annotators, Separable in Judge LLMs 2026

Chun, Y. (2026). Moral Orientation and Calibration: Coupled in Human Annotators, Separable in Judge LLMs. *ICML 2026 Pluralistic Alignment Workshop* (accepted). Paper · Code

When Fear Becomes Beauty: Genre as Context in Human and AI Aesthetic Judgment 2026

Wang, I. W., & **Chun, Y.** (2026). Manuscript in preparation. A technical framework for contextual value alignment of large language model evaluators in the aesthetic domain, extending the orientation and calibration decomposition to genre-typed cultural contexts. Cross-institutional collaboration between the Institute for Ethics in AI (TU Munich) and KT Frontier AI Lab.

A Smile Blinds the Judge: Role-Blind Valence Pooling in Vision-Language Models 2026

Wang, I. W., & **Chun, Y.** (2026). Manuscript in preparation. Vision-language safety judges pool facial valence without binding emotion to narrative roles: a victim's smile lowers judged danger, while a perpetrator's smile lowers judged malice. The effect remains significant from 3B local models to a 397B frontier model, indicating that scale alone does not supply the missing role binding.

A Hybrid Corpus-to-Taxonomy Pipeline for Physical AI Risk Cards 2026

Chun, Y. (2026). Technical Report v1.0, Responsible AI Department, Frontier AI Lab, KT Corporation. Documents the data collection, deduplication and hierarchical assignment pipeline behind the Physical AI Risk Taxonomy, with reproducibility artefacts.

AI Technology Specialization and National Competitiveness 2024

Chun, Y., Hur, J., & Hwang, J. (2024). *PLOS ONE*. Link

The Nexus of Artificial Intelligence and Green Innovation: A Cross-Density Analysis at the Country Level 2024

Chun, Y., & Hwang, J. (2024). *Journal of the Knowledge Economy*. Link

The Moderating Role of Design Innovation in the Relationship Between Technology Complexity and Firm Performance 2024

Hur, J., **Chun, Y.**, Hwang, J., & Kim, K. (2024). *Technology Analysis & Strategic Management*, 1–14. Link

Selected Projects

Physical AI Risk Taxonomy 2026 – Present

Project Lead, Frontier AI Lab, KT (Korea). A public, bilingual evidence map of risk cards for AI systems that perceive, decide and act in the physical world, from robots and humanoids to drones and autonomous

vehicles. Built with a hybrid method that pairs a human-defined risk hierarchy with bottom-up discovery from a large corpus of papers, patents and policy documents, and released with alignment tags, audit logs and a public technical report. Live site · Repository

Physical AI Risk Assessment for Urban Deployment · with MIT Senseable City Lab Aug 2026 – Present

Project Lead, Frontier AI Lab, KT (Korea), with the MIT Senseable City Lab and the City of Seoul. A twelve-month project that applies the Physical AI Risk Taxonomy to real urban settings, building a risk-assessment framework for how embodied AI and street-level conditions interact in a city, and working towards a Korea-specific evaluation standard.

Cognitive Security of Hybrid Human–AI Systems 2026 – Present

Research programme in development examining when AI acts as a cognitive complement rather than a substitute. It models the human–AI ensemble as a directed weighted delegation graph and defines autonomy as the persistence of pathways that bypass the AI subsystem, hypothesising a critical, percolation-like transition at which autonomy collapses while aggregate performance still appears intact.

Red-Teaming and Defence of Retrieval-Augmented Generation (RAG) Systems Oct 2025 – Nov 2026

Project lead at Frontier AI Lab, KT (Responsible AI Department), with Korea University (Prof. Buru Chang's group). Black-box red-teaming of agentic RAG systems (the KidnapRAG attack, accepted to EMNLP 2026 Main Conference) and corresponding defences, delivering the ARS-Bench security benchmark, an open-source toolkit and a registered patent, and connecting adversarial security research to the responsible deployment of B2B RAG services.

Contextual Value Alignment through Genre-Based Aesthetic Judgement 2026 – Present

International collaboration with Isa Weisha Wang (Institute for Ethics in AI, Technical University of Munich). A framework for measuring where large language model evaluators diverge from human interpretive communities in aesthetic judgement, using film genres as measurable cultural contexts. It distinguishes universal from context-dependent value dimensions and locates human and AI gaps through a sign-flip test and a subspace-resolved orientation and calibration audit. Companion to the orientation and calibration work, with a joint manuscript in preparation.

AI Topic Space: Policy, Science and Technology in the Korean AI Knowledge Ecosystem 2025 – Present

Principal investigator, with STEPI and KDI (Korea). An interactive semantic map of AI topics across policy reports, papers and patents that lets users explore how the science, technology and policy agendas align and where Korean policy falls behind. Live site · Repository · HuggingFace dataset

DCI Patent Atlas 2026

A reproducible patent atlas spanning a seven-position Data Center Interconnect value chain from materials through advanced packaging to systems and submarine operations. Its citation network contains 2,188,730 directed edges over 927,481 applications; a 35-country, 2003–2025 panel links DCI capability to comparative advantage in 2.70 million AI-science papers and AI technology. DCI capability transfers directly into adjacent AI patent fields, but entry into AI science requires an existing scientific base; advanced packaging matters on both margins. Outputs include a public site, repository, technical report and panel paper. Site · Repository

PATSTAT AI Complete Master, 1950–2026 2026

A research-ready dataset of 2,330,553 unique AI patent applications and 175 variables, partitioned by priority-year period and released with processing code, quality metadata and checksums. An accompanying reproducible collector notebook automates six relational query groups against PATSTAT Online. HuggingFace dataset · Collector

Innovation Characteristics of Generative AI and Large Language Models: A Patent Analytics Approach Jul 2025 – Jan 2026

Research Collaborator, KT Frontier AI Lab, with STEPI. Patent embeddings and ensemble modelling assessed dominant-design emergence and cross-national competition.

Analysis of Industrial AI Innovation and National Competitiveness: A Scientometric Approach Dec 2025 – Jan 2026

Research Collaborator, Seoul National University and KDI. Patent analytics and revealed comparative advantage were used to examine domain specialisation and industrial AI competitiveness.

Patents

Method and Apparatus for Automatically Constructing a Physical AI Risk Taxonomy Filed Aug 2026

Chun, Y. et al. (2026) [P20260372]. Korean patent application (filed 3 Aug 2026) covering the automated construction of a risk classification system for Physical AI, arising from the Physical AI Risk Taxonomy project.

Method, Server and Computer Program for Dynamically Adjusting Value Information 2025 – Present

Chun, Y., Park, Y., Yoon, J., & Lee, S. (2025) [P20250217]. A value-alignment method for generative AI combining multi-agent debate and reinforcement-learning-based reward modelling to align dynamically with evolving ethical and social values without continuous human intervention.

KidnapRAG: Black-Box Attack for Hijacking Reasoning in Agentic RAG Systems Registered Jul 2026

Chun, Y. (first inventor), Choi, C., Kim, E., Lee, K., Jeong, J., Kim, E., Oh, M., Jang, J., & Chang, B. (2026). Registered Korean patent (Jul 2026) on black-box poisoning attacks against agentic retrieval-augmented generation systems, with Chun as first inventor.

Honours and Awards

- **Travel Grant Winner**, The Responsible AI Forum (TRAIF), Technical University of Munich · 2026.
- **Excellent Patent Award**, Invention Competition 2025, KT Corporation R&D Center · Aug 2025 (patent *Method, Server and Computer Program for Dynamically Adjusting Value Information*).
- **Best PhD Paper Award**, KOSIME 2024, Korea Society of Innovation, Management, and Economics · Jul 2024.

Conferences and Seminars

- **NRC Outcomes Forum, "The AI Transition of Industry and Social Institutions: Inclusive Policy Design and Response Strategies"**, Brahms Hall, Hotel President, Seoul (7 Oct 2026) · invited Thematic Presentation 3, on policy-gap visualisation, at the dissemination forum of the NRC national AI collaborative research programme.
- **OECD, Paris** (1–2 Oct 2026) · Invited presenter with the KDI delegation, alongside colleagues from the Bank of Korea, Chung-Ang University and the KDI School. Gave two presentations: policy-gap visualisation at the working meeting with the STI AI and Emerging Digital Technologies Division on AI risk classification (1 Oct), and an analysis of Korea's AI policy landscape at the KDI–OECD Expert Roundtable on AI-Driven Innovation, Technology Diffusion and Productivity, convened with the STI Productivity, Innovation and Entrepreneurship Division (2 Oct).
- **Funding at the Frontier — South Korea**, British Embassy Seoul and the UK Advanced Research and Invention Agency (ARIA), Aston Hall, Seoul (19 Aug 2026) · invited participant.
- **Korean Association for Policy Studies (KAPS) 2026 Summer Conference**, Gwangju, Korea (14 Aug 2026) · "Policy-Gap Visualisation," in the STEPI-convened session *Inclusiveness and Policy-Agenda Gaps in the AI Transition: linkage mechanisms, visualisation and discourse*, within the NRC national AI collaborative research programme.
- **KDI Expert Advisory Meeting on AI Acceptability**, Sejong, Korea (4 Aug 2026) · invited presentation of the policy-gap visualisation results, within the NRC national AI collaborative research programme.
- **ICML 2026 Pluralistic Alignment Workshop** (Poster Session), 2026 · "Moral Orientation and Calibration: Coupled in Human Annotators, Separable in Judge LLMs."
- **2024 Seminar on Artificial Intelligence in Budapest**, Hungary (Oct 2024) · "Strategic Approaches to AI Development and Implementation."
- **KOSIME 2024**, Jeju, Korea (Jul 2024) · "National AI Capabilities and Product Diversification: A Multidimensional Knowledge Space Approach."
- **6th Geography of Innovation Conference (GEOINNO 2022)**, Milan, Italy (Jul 2022) · "Economic Resilience of EU Regions to Localized Attacks."
- **ACM SIGCHI 2020 (Fair & Responsible AI Workshop)**, Hawaii (Apr 2020) · "How Does

Perceived Fairness Differ? Exploring Multicultural Landscape for Ethical AI Design."

- **The Future Use of Autonomous Vehicles** (Invited Seminar), Paris/EHESS (Jun 2019) · "Technology Shock, Microdynamics, and Industrial Leadership Change."

Professional Service

Journal Reviewer

2023 – Present

Scientific Reports, *PLOS ONE*, *Journal of the Knowledge Economy*, *European Research on Management and Business Economics*, *Business Strategy and the Environment*, *Research in Globalization*, and *Humanities and Social Sciences Communications*.

AI Governance & Innovation Policy Consulting

2024 – Present

Invited Speaker, KDI (Nov 2024); AI Industry Roundtable, Seoul AI Hub (Aug 2024); AI patent/paper data analysis, KDI (Oct 2024); AI regulation research, STEPI (Jun 2024); AI-industry patent analysis, STEPI (Apr 2024).

Expert Panellist, Seoul AI Ethics Index

2026

Seoul AI Foundation. One of fifteen experts in the two-round Delphi study establishing the city's AI ethics index.

Professional Affiliations & Networking

2024 – Present

GSMA AI for Impact Taskforce · Responsible AI Maturity Roadmap; standardised benchmarks across five key dimensions.

Policy and Industry Advisory

2023 – Present

Expert Advisor, HYGINO (carbon-footprint policy design); Advisor, National Grand Challenges Energy Division, STEPI; Advisory Committee & Author, KOTRA Global Economic and Trade Outlook Report.

Teaching Experience

- **Lecturer** · **Business Analytics**, Graduate School of Technology and Innovation Management, UNIST (Aug 2024 – Present): hands-on data analysis in Python/R, AI and generative-AI foundations, group-project supervision.
- **Invited Lecturer** · **Generative and Agentic AI: Innovation Mechanisms and Ethics**, Pukyong National University (Mar 2026): the evolutionary trajectory of AI from perception to generative, agentic, and physical AI; agentic AI mechanisms (multimodal interaction, reasoning/planning, action execution, self-learning); and the ethics and regulatory landscape of generative AI.
- **Invited Lecturer** · **AI Co-evolution and Innovation Dynamics in Mobile Technology**, Underwood International College, Yonsei University (Nov 2025): evolution of mobile/communication technologies from 1G to 5G/6G, energy and security challenges.
- **Invited Lecturer** · **Generative and Agentic AI: Innovation Mechanisms and Ethics**, Kyung Hee University (Mar 2025).

Skills

Programming: Python (ML, data science, NLP), SQL, R, LaTeX.
AI & NLP: Sentence-BERT, PatentSBERTa, RAG systems.
Analytics: network analysis, embeddings, clustering, UMAP, scientometrics, patent analytics.
Statistics & visualisation: STATA, Tableau.

Language Skills

Korean · native. **English** · C1/C2 (Listening C2, Reading C2, Spoken C1, Writing C2). **Japanese** · B2/C1.

References

- **Prof. Chris Russell** · Dieter Schwarz Associate Professor in AI, Government and Policy, University of Oxford (Oxford Internet Institute). chris.russell@oii.ox.ac.uk
- **Prof. Christoph Lütge** · Professor of Business Ethics; Director, Institute for Ethics in AI (IEAI), Technical University of Munich. christoph.luetge@tum.de
- **Prof. Junseok Hwang** · Professor, TEMEP, Seoul National University; MSc and PhD supervisor. junhwang@snu.ac.kr
- **Dr Jongrip Kim** · Team Leader, Research Impact and Dissemination Team, Science and Technology Policy Institute (STEPI); principal investigator of the NRC national AI collaborative research programme. lipk@stepi.re.kr
- **Prof. Keungoui Kim** · Professor, Underwood International College, Yonsei University. awekim@yonsei.ac.kr
- **Prof. Camille Salinesi** · University Professor, Université Paris 1 Panthéon-Sorbonne; Co-Director, Observatoire de l'IA. camille.salinesi@univ-paris1.fr